

# Yash Mishra

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## Professional Summary

Senior Machine Learning Engineer with experience in developing and deploying **ML, Generative AI and custom vision solutions**. Expertise in **LLMs, Multimodal RAG, Predictive modeling, MLOps** and **cloud platforms** like **GCP**. Proven success in building AI-powered applications for healthcare, e-commerce and manufacturing industry with innovative problem-solving and leadership skills.

## Experience

**Senior Machine Learning Engineer** | Quantiphi Analytics Pvt. Ltd., Bangalore

*Aug 2022 – Present*

- Developed and deployed **ML models** for predictive analytics and Generative AI.
- Integrated **LLMs** for advanced solutions in healthcare and e-commerce domains.
- **Led teams of data scientists**, improving project delivery and client satisfaction.
- Developed **production-ready CICD MLOps pipelines** on **GCP** to boost operational efficiency for healthcare providers.
- Designed and implemented **custom ML solutions** for various industries.
- Built predictive analytics models, data preprocessing pipelines, and deployed solutions at scale.
- Delivered **Generative AI** projects, including LLM-driven applications.
- Worked on **sensor data** of manufacturing industry
- Built an advanced custom vision model to detect defects in the chipset manufacturing.

## Technical Skills

- **Languages:** Python, SQL, Bash
- **Machine Learning & Deep Learning:** Supervised & Unsupervised Learning, Generative AI, LLMs, Prompt Engineering, RAG, Transformers, Vision-Language Models, Object Detection, Image Segmentation, Deep Learning (CNN, RNN, U-Net, U-Net++), XGBoost, AutoML, SHAP, A/B Testing, Time-Series Forecasting (ARIMA, LSTM)
- **NLP & LLM Ecosystem:** Gemini 1.5 Pro, OpenAI GPT, LangChain, Hugging Face Transformers, Vector DBs (FAISS, Pinecone), Token Optimization, Few-shot/Zero-shot Prompting, PII/PHI Masking (Google DLP), Text-to-Image Embeddings, Retrieval-Augmented Generation (RAG), Gemini Vision
- **MLOps & Model Deployment:** Vertex AI Pipelines, Kubeflow, TensorFlow Extended (TFX), ML Metadata Tracking, Vertex AI Model Monitoring, TensorFlow Serving, Docker, CI/CD (Cloud Build, GitHub Actions), Feature Store, Model Registry
- **Cloud Platforms:** Google Cloud Platform (Vertex AI, BigQuery, Cloud Functions, Cloud Scheduler, AutoML, Looker, GCS), AWS (S3, EC2 – basic exposure)
- **Data Engineering & Visualization:** Pandas, NumPy, Apache Beam (basics), Spark (basic exposure), Data Validation, Looker, BigQuery, Matplotlib, Seaborn, SQL Optimization
- **Tools & Frameworks:** TensorFlow, Keras, Scikit-learn, OpenCV, PyTorch (basic), Git, LabelMe, Streamlit, FastAPI, Docker, Airflow (basics)
- **Dev & Experiment Tracking:** Git, DVC, Weights & Biases (W&B), MLflow (explored)

## Key Projects

### Chipset Defect Detection using Image Segmentation (U-Net & U-Net++)

- Engineered segmentation masks using custom class mapping and label encoding for multi-class defect detection.
- Trained and fine-tuned U-Net and U-Net++ architectures using Keras and TensorFlow, achieving precise localization of micro-defects.
- Applied advanced data augmentation and optimized IoU and Dice loss metrics for accurate segmentation of small defects.
- Enabled automation of visual QA, reducing manual inspection effort and enhancing defect traceability.

### Multimodal, E-commerce product recommendations(LLM + RAG)

- Used Gemini 1.5 Pro for image-to-text generation and contextual product description creation.
- Engineered multimodal embeddings combining CLIP-based visual and BERT-based text vectors for personalized retrieval.
- Implemented Retrieval-Augmented Generation (RAG) pipeline for product recommendations based on user queries.
- Deployed solution on GCP Vertex AI and integrated with e-commerce platform APIs for real-time product suggestions.

### Sensor Data Analysis for Downtime Reduction, (Manufacturing Data Engineering)

- Analyzed 1.2M+ time-series sensor logs to identify downtime patterns in manufacturing equipment (rolling mill, furnace).
- Selected 96 out of 1600 tags using SHAP-based feature selection and correlation heatmaps.
- Implemented alert-based logic and Looker dashboards powered by BigQuery to notify potential machine failures proactively.

**Medical Entity Extraction & Clinical Prediction (LLM: Gemini 1.5 Pro)**

- Built an LLM-based data pipeline using Gemini 1.5 Pro for extracting clinical entities from raw physician notes.
- Used dynamic token shortening and async batching to optimize prompt inference latency.
- Masked PHI/PII using Google DLP and formatted structured outputs into HL7-compatible schema for EHR ingestion.

**AKI Risk Prediction using XGBoost, AutoML, and BQML**

- Benchmarked 10 ML models on GCP for Acute Kidney Injury (AKI) prediction using clinical records.
- Trained and evaluated XGBoost, AutoML Tables, and BigQuery ML with explainability using SHAP and AUC/F1 as metrics.
- Deployed models via Vertex AI Pipelines with a focus on reproducibility and monitoring performance post-deployment.

**MLOps automation for healthcare, predictive models, Gen AI engine (GCP)**

- Designed CI/CD ML workflows using Vertex AI Pipelines, Kubeflow components, and Cloud Scheduler.
- Used TFX and Data Validation for preprocessing, schema checks, and transformation consistency across pipeline stages.
- Enabled container-based deployment with Cloud Build and Artifact Registry, reducing model iteration time by 40%.
- Implemented drift detection and latency monitoring using Vertex AI Model Monitoring and custom alerting hooks.

**Video stitching and object detection for broadcast channel through Custom YOLO V8 model and Open CV SIFT**

- Performed real time video stitching to produce a streamable broadcast ready video from 7 cameras placed in different angles and applied object detection to detect player and ball movement.

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**Key Impact Metrics in Experience**

- Reduced human manual defect detection time by ~98% and achieved 93%+ accuracy on defect detection by optimizing the U-Net model for micro-defect segmentation using custom augmentations.
- Reduced model training and deployment cycle time by 40% via automated CI/CD pipelines using Vertex AI, TFX, and Cloud Build.
- Cut manual medical note review time by 80% through structured output generation using LLM-based entity extraction pipeline.
- Achieved 91%+ AUC in AKI risk prediction benchmark, outperforming baseline models and enabling faster triage decisions.
- Minimized unplanned machine downtime by 15–20% by identifying key sensor thresholds and visualizing alerts on Looker.

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**Education**

**B.Tech (Hons.)**| Lakshmi Narain College of Technology, Bhopal  
2018 – 2022

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**Certifications**

- Google certified **Generative AI Leader** ([Click here to view certificate](#))
- Google Certified **Professional ML Engineer** (GCP) ([Click here to view certificate](#))
- Google Certified **Associate Cloud Engineer** ([Click here to view certificate](#))
- 17 GenAI Badges by **Google**
- **L400 - Build and deploy Generative AI solution using RAG** ([Click here to view certificate](#))
- **Unsung Hero Award, Quantiphi** — Recognized for consistently delivering high-impact ML and Generative AI solutions across healthcare and manufacturing domains.
- **SPOT Award, Quantiphi** — Awarded for outstanding contribution in automating medical entity extraction and clinical risk prediction using Gemini 1.5 Pro.

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**Leadership & Mentorship**

- Led a team of 3–4 data scientists across multiple ML and GenAI projects, guiding sprint planning, code reviews, and deployment practices.
- Mentored new joiners and interns on topics like LLMOps, image segmentation workflows, and GCP-based model deployment.
- Took ownership of end-to-end solution delivery, serving as the client SPOC for high-priority healthcare and manufacturing engagements.
- Spearheaded internal knowledge-sharing sessions on Vertex AI Pipelines, Prompt Optimization, and Gemini-based RAG architecture.

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**Hobbies**

- Market Analysis (NSE, Cryptocurrencies) | Traveling | Reading
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